

μOS++ III SMP

AArch32 and AArch64 test commands, one by one

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Every QEMU and hardware test of the AArch32 and AArch64 platforms, written as the explicit commands that `test_smpl/run-qemu.sh` and `test_smpl/run-hw.sh` (through each board's `hw.sh`) execute, without the scripts.

Platform	Port	QEMU tests	Hardware tests
<code>aarch32-rpi-zero-2w</code>	<code>micro-os-plus-iii-aarch32</code>	15	15
<code>aarch32-rpi3b</code>	<code>micro-os-plus-iii-aarch32</code>	15	14
<code>aarch64-rpi-zero-2w</code>	<code>micro-os-plus-iii-aarch64</code>	15	15
<code>aarch64-rpi3b</code>	<code>micro-os-plus-iii-aarch64</code>	15	14
<code>aarch32-luckfox-lyra</code>	<code>micro-os-plus-iii-aarch32</code>	—	20

The addresses (`__smp_spin`) in section 4 were read on 2026-10-09 from the **debug** builds of the five platforms (`*-cmake-gcc-debug`) made from:

Repository	Branch	Commit
<code>micro-os-plus-iii</code>	<code>smp</code>	<code>d780c76c</code>
<code>micro-os-plus-iii-aarch32</code>	<code>smp</code>	<code>e721dea</code>
<code>micro-os-plus-iii-aarch64</code>	<code>smp</code>	<code>8151c26</code>

They change whenever a test is built differently (release, another commit); section 4.1 shows how to read them again.

1. Paths and tools

```
WORK=/tmp
K=$WORK/micro-os-plus-iii           # kernel
A32=$WORK/micro-os-plus-iii-aarch32 # AArch32 port
A64=$WORK/micro-os-plus-iii-aarch64 # AArch64 port
BUILD=$K/tests/build

QEMU=$HOME/.local/xPacks/@xpack-dev-tools/qemu-arm/9.2.4-1.1/.content/bin/qemu-system-aarch64
OPENOCD=$HOME/.local/xPacks/@xpack-dev-tools/openocd/0.12.0-7.1/.content/bin/openocd
SCRIPTS=$HOME/.local/xPacks/@xpack-dev-tools/openocd/0.12.0-7.1/.content/openocd/scripts
TC32=$HOME/.local/xPacks/@xpack-dev-tools/arm-none-eabi-gcc/15.2.1-1.1.1/.content/bin
TC64=$HOME/.local/xPacks/@xpack-dev-tools/aarch64-none-elf-gcc/15.2.1-1.1.1/.content/bin
```

These are the versions the scripts pick on this machine: the newest `qemu-arm` and `openocd` xPacks installed, and the 15.x toolchains.

2. Install and build

```
cd $A32 && xpm install && xpm link
cd $A64 && xpm install && xpm link

cd $K/tests
for c in aarch32-rpi-zero-2w aarch32-rpi3b aarch64-rpi-zero-2w aarch64-rpi3b aarch32-luckfox-lyra; do
    xpm install --config $c-cmake-gcc-debug
    xpm run prepare --config $c-cmake-gcc-debug
    xpm run build --config $c-cmake-gcc-debug
done
```

(Replace `-debug` by `-release` for the release images.) The images are in `$BUILD/<platform>-cmake-gcc-debug/platform-bin/port-tests/test/`:

File	Used by
<code><test>-qemu.bin</code>	QEMU (raw image)
<code>shim8.img</code>	QEMU, AArch32 only (the AArch64-to-AArch32 boot shim)
<code><test>-hwd</code>	hardware (the ELF; a link to <code><test>-hwd.elf</code>)

3. QEMU

3.1 The command

AArch32 — QEMU's `raspi3b` starts its cores in AArch64, so it boots `shim8.img`, which drops to AArch32 and jumps to the test image loaded at `0x10000`:

```
timeout <seconds> $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
[-drive file=<card image>,if=sd,format=raw] \  
-kernel $D/shim8.img -device loader,file=$D/<test>-qemu.bin,addr=0x10000 \  
< /dev/null
```

AArch64 — the test image is the kernel:

```
timeout <seconds> $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
[-drive file=<card image>,if=sd,format=raw] \  
-kernel $D/<test>-qemu.bin \  
< /dev/null
```

The test passed when its output has a line `RESULT: PASS`. `RESULT: SKIP` is a skip (`usb_test`: QEMU has no USB device mode). `RESULT: FAIL`, no `RESULT` line, or exit status 124 (the `timeout` expired) is a failure.

3.2 The SD card image

Four tests need a card: a fresh image before the run, deleted after it.

```
# sd_test: a card seeded with a flatfs volume, by the test's own tool  
python3 <port>/test/<board>/sd_test/flatfs_tool.py make-seed /tmp/sd_test.disk.img  
  
# smp-mat-sdcard-test, smp-num-test, smp-pipeline-test: a blank 4 GiB sparse file  
truncate -s 4G /tmp/<test>.disk.img  
  
rm -f /tmp/<test>.disk.img      # after the run
```

Checked on 2026-10-09 with these exact commands: AArch32 and AArch64 `smp_test0` (`RESULT: PASS (10 heartbeats on core 0)`, exit 0), and AArch32 `sd_test` with a seeded card (`RESULT: PASS (SD + flatfs verified)`, exit 0).

3.3 aarch32-rpi-zero-2w — QEMU

```
D=$BUILD/aarch32-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test
```

cmsis-os-validator

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/cmsis-os-validator-qemu.bin,addr=0x10000 \  
< /dev/null
```

mutex-stress

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/mutex-stress-qemu.bin,addr=0x10000 \  
< /dev/null
```

rtos-apis

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/rtos-apis-qemu.bin,addr=0x10000 \  
< /dev/null
```

sd_test

```
python3 $A32/test/rpi-zero-2w/sd_test/flatfs_tool.py make-seed /tmp/sd_test.disk.img  
timeout 450 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-drive file=/tmp/sd_test.disk.img,if=sd,format=raw \  
-kernel $D/shim8.img -device loader,file=$D/sd_test-qemu.bin,addr=0x10000 \  
< /dev/null  
rm -f /tmp/sd_test.disk.img
```

smp-mat-sdcard-test

```
truncate -s 4G /tmp/smp-mat-sdcard-test.disk.img
timeout 2000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-mat-sdcard-test.disk.img,if=sd,format=raw \
    -kernel $D/shim8.img -device loader,file=$D/smp-mat-sdcard-test-qemu.bin,addr=0x10000 \
    < /dev/null
rm -f /tmp/smp-mat-sdcard-test.disk.img
```

smp-mat-test

```
timeout 900 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -kernel $D/shim8.img -device loader,file=$D/smp-mat-test-qemu.bin,addr=0x10000 \
    < /dev/null
```

smp-num-test

```
truncate -s 4G /tmp/smp-num-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-num-test.disk.img,if=sd,format=raw \
    -kernel $D/shim8.img -device loader,file=$D/smp-num-test-qemu.bin,addr=0x10000 \
    < /dev/null
rm -f /tmp/smp-num-test.disk.img
```

smp-pipeline-test

```
truncate -s 4G /tmp/smp-pipeline-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-pipeline-test.disk.img,if=sd,format=raw \
    -kernel $D/shim8.img -device loader,file=$D/smp-pipeline-test-qemu.bin,addr=0x10000 \
    < /dev/null
rm -f /tmp/smp-pipeline-test.disk.img
```

smp-pro-cons-test

```
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -kernel $D/shim8.img -device loader,file=$D/smp-pro-cons-test-qemu.bin,addr=0x10000 \
    < /dev/null
```

smp_test0

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test0-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test1

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test1-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test2

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test2-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test3

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test3-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test4

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test4-qemu.bin,addr=0x10000 \  
< /dev/null
```

usb_test

```
timeout 200 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/usb_test-qemu.bin,addr=0x10000 \  
< /dev/null
```

3.4 aarch32-rpi3b — QEMU

```
D=$BUILD/aarch32-rpi3b-cmake-gcc-debug/platform-bin/port-tests/test
```

cmsis-os-validator

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/cmsis-os-validator-qemu.bin,addr=0x10000 \  
< /dev/null
```

mutex-stress

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/mutex-stress-qemu.bin,addr=0x10000 \  
< /dev/null
```

rtos-apis

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/rtos-apis-qemu.bin,addr=0x10000 \  
< /dev/null
```

sd_test

```
python3 $A32/test/rpi3b/sd_test/flatfs_tool.py make-seed /tmp/sd_test.disk.img  
timeout 450 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-drive file=/tmp/sd_test.disk.img,if=sd,format=raw \  
-kernel $D/shim8.img -device loader,file=$D/sd_test-qemu.bin,addr=0x10000 \  
< /dev/null  
rm -f /tmp/sd_test.disk.img
```

smp-mat-sdcard-test

```
truncate -s 4G /tmp/smp-mat-sdcard-test.disk.img
timeout 2000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-mat-sdcard-test.disk.img,if=sd,format=raw \
    -kernel $D/shim8.img -device loader,file=$D/smp-mat-sdcard-test-qemu.bin,addr=0x10000 \
    < /dev/null
rm -f /tmp/smp-mat-sdcard-test.disk.img
```

smp-mat-test

```
timeout 900 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -kernel $D/shim8.img -device loader,file=$D/smp-mat-test-qemu.bin,addr=0x10000 \
    < /dev/null
```

smp-num-test

```
truncate -s 4G /tmp/smp-num-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-num-test.disk.img,if=sd,format=raw \
    -kernel $D/shim8.img -device loader,file=$D/smp-num-test-qemu.bin,addr=0x10000 \
    < /dev/null
rm -f /tmp/smp-num-test.disk.img
```

smp-pipeline-test

```
truncate -s 4G /tmp/smp-pipeline-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-pipeline-test.disk.img,if=sd,format=raw \
    -kernel $D/shim8.img -device loader,file=$D/smp-pipeline-test-qemu.bin,addr=0x10000 \
    < /dev/null
rm -f /tmp/smp-pipeline-test.disk.img
```

smp-pro-cons-test

```
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -kernel $D/shim8.img -device loader,file=$D/smp-pro-cons-test-qemu.bin,addr=0x10000 \
    < /dev/null
```

smp_test0

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test0-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test1

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test1-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test2

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test2-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test3

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test3-qemu.bin,addr=0x10000 \  
< /dev/null
```

smp_test4

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/smp_test4-qemu.bin,addr=0x10000 \  
< /dev/null
```

usb_test

```
timeout 200 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/shim8.img -device loader,file=$D/usb_test-qemu.bin,addr=0x10000 \  
< /dev/null
```

3.5 aarch64-rpi-zero-2w — QEMU

```
D=$BUILD/aarch64-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test
```

cmsis-os-validator

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/cmsis-os-validator-qemu.bin \  
< /dev/null
```

mutex-stress

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/mutex-stress-qemu.bin \  
< /dev/null
```

rtos-apis

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/rtos-apis-qemu.bin \  
< /dev/null
```

sd_test

```
python3 $A64/test/rpi-zero-2w/sd_test/flatfs_tool.py make-seed /tmp/sd_test.disk.img  
timeout 450 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-drive file=/tmp/sd_test.disk.img,if=sd,format=raw \  
-kernel $D/sd_test-qemu.bin \  
< /dev/null  
rm -f /tmp/sd_test.disk.img
```

smp-mat-sdcard-test

```
truncate -s 4G /tmp/smp-mat-sdcard-test.disk.img
timeout 2000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-mat-sdcard-test.disk.img,if=sd,format=raw \
    -kernel $D/smp-mat-sdcard-test-qemu.bin \
    < /dev/null
rm -f /tmp/smp-mat-sdcard-test.disk.img
```

smp-mat-test

```
timeout 900 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -kernel $D/smp-mat-test-qemu.bin \
    < /dev/null
```

smp-num-test

```
truncate -s 4G /tmp/smp-num-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-num-test.disk.img,if=sd,format=raw \
    -kernel $D/smp-num-test-qemu.bin \
    < /dev/null
rm -f /tmp/smp-num-test.disk.img
```

smp-pipeline-test

```
truncate -s 4G /tmp/smp-pipeline-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -drive file=/tmp/smp-pipeline-test.disk.img,if=sd,format=raw \
    -kernel $D/smp-pipeline-test-qemu.bin \
    < /dev/null
rm -f /tmp/smp-pipeline-test.disk.img
```

smp-pro-cons-test

```
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
    -semihosting-config enable=on,target=native \
    -kernel $D/smp-pro-cons-test-qemu.bin \
    < /dev/null
```

smp_test0

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test0-qemu.bin \  
< /dev/null
```

smp_test1

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test1-qemu.bin \  
< /dev/null
```

smp_test2

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test2-qemu.bin \  
< /dev/null
```

smp_test3

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test3-qemu.bin \  
< /dev/null
```

smp_test4

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test4-qemu.bin \  
< /dev/null
```

usb_test

```
timeout 200 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/usb_test-qemu.bin \  
< /dev/null
```

3.6 aarch64-rpi3b — QEMU

```
D=$BUILD/aarch64-rpi3b-cmake-gcc-debug/platform-bin/port-tests/test
```

cmsis-os-validator

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/cmsis-os-validator-qemu.bin \  
< /dev/null
```

mutex-stress

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/mutex-stress-qemu.bin \  
< /dev/null
```

rtos-apis

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/rtos-apis-qemu.bin \  
< /dev/null
```

sd_test

```
python3 $A64/test/rpi3b/sd_test/flatfs_tool.py make-seed /tmp/sd_test.disk.img  
timeout 450 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-drive file=/tmp/sd_test.disk.img,if=sd,format=raw \  
-kernel $D/sd_test-qemu.bin \  
< /dev/null  
rm -f /tmp/sd_test.disk.img
```

smp-mat-sdcard-test

```
truncate -s 4G /tmp/smp-mat-sdcard-test.disk.img
timeout 2000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
  -semihosting-config enable=on,target=native \
  -drive file=/tmp/smp-mat-sdcard-test.disk.img,if=sd,format=raw \
  -kernel $D/smp-mat-sdcard-test-qemu.bin \
  < /dev/null
rm -f /tmp/smp-mat-sdcard-test.disk.img
```

smp-mat-test

```
timeout 900 $QEMU -M raspi3b -smp 4 -nographic -serial none \
  -semihosting-config enable=on,target=native \
  -kernel $D/smp-mat-test-qemu.bin \
  < /dev/null
```

smp-num-test

```
truncate -s 4G /tmp/smp-num-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
  -semihosting-config enable=on,target=native \
  -drive file=/tmp/smp-num-test.disk.img,if=sd,format=raw \
  -kernel $D/smp-num-test-qemu.bin \
  < /dev/null
rm -f /tmp/smp-num-test.disk.img
```

smp-pipeline-test

```
truncate -s 4G /tmp/smp-pipeline-test.disk.img
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
  -semihosting-config enable=on,target=native \
  -drive file=/tmp/smp-pipeline-test.disk.img,if=sd,format=raw \
  -kernel $D/smp-pipeline-test-qemu.bin \
  < /dev/null
rm -f /tmp/smp-pipeline-test.disk.img
```

smp-pro-cons-test

```
timeout 1000 $QEMU -M raspi3b -smp 4 -nographic -serial none \
  -semihosting-config enable=on,target=native \
  -kernel $D/smp-pro-cons-test-qemu.bin \
  < /dev/null
```

smp_test0

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test0-qemu.bin \  
< /dev/null
```

smp_test1

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test1-qemu.bin \  
< /dev/null
```

smp_test2

```
timeout 300 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test2-qemu.bin \  
< /dev/null
```

smp_test3

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test3-qemu.bin \  
< /dev/null
```

smp_test4

```
timeout 150 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/smp_test4-qemu.bin \  
< /dev/null
```

usb_test

```
timeout 200 $QEMU -M raspi3b -smp 4 -nographic -serial none \  
-semihosting-config enable=on,target=native \  
-kernel $D/usb_test-qemu.bin \  
< /dev/null
```

4. Hardware

4.1 What a hardware run does

One test per power cycle: the image is loaded into RAM over whatever the previous test left there, and neither board has a reset a script can drive. Power-cycle the board before each test. These commands do not use the UART console; keep your own terminal on it (for example `tio -b 115200 /dev/ttyACM0`). The test's semihosting output appears in the OpenOCD output; the test passed when it prints `RESULT: PASS`. Then stop OpenOCD (Ctrl-C).

The session file is written with a quoted here-document (so that the Tcl `$score` stays literal); the ELF path is then put in with `sed`.

The two values in each test's session come from its ELF:

```
$TC32/arm-none-eabi-readelf -h $D/<test>-hwd | awk '/Entry point/{print $NF}' # entry
$TC32/arm-none-eabi-nm $D/<test>-hwd | awk '$3 == "__smp_spin" {print $1}' # __smp_spin
# AArch64: $TC64/aarch64-none-elf-readelf and $TC64/aarch64-none-elf-nm
```

`__smp_spin` is zeroed after the load: 4 32-bit words on AArch32 (`uint32_t[4]`), 8 on AArch64 (`uint64_t[4]`).

4.2 Raspberry Pi (Zero 2 W and 3 B), both ports

	AArch32	AArch64
OpenOCD config, J-Link (default)	<code>\$A32/test/boards/rpi-zero-2w/openocd-jlink-rpi3.cfg</code>	<code>\$A64/test/boards/rpi-zero-2w/openocd-jlink-rpi3.cfg</code>
OpenOCD config, Olimex ARM-USB-OCD	<code>\$A32/test/boards/rpi-zero-2w/openocd-olimex.cfg</code>	<code>\$A64/test/boards/rpi-zero-2w/openocd-olimex.cfg</code>
<code>-c init</code>	no (the config runs <code>init</code> itself)	yes (the config is declarative)
entry	<code>0x1003c</code>	<code>0x80000</code>
resume	set CPSR to <code>0x600001da</code> , then resume at the entry	set PC to the entry, then resume

The Pi 3 B uses the Zero 2 W's configs (same BCM2837). Each test is three steps:

1. **Reset** through the watchdog (`PM_RSTC` / `PM_WDOG` at `0x3f10001c` and `0x3f100024`), then wait 12 s for the board to boot.
2. **Write the session** (`/tmp/session.tcl`): halt the four cores, enable semihosting on each, load the ELF, zero `__smp_spin`, resume the cores.
3. **Run OpenOCD** with the config and the session.

The sections below write every test with the J-Link config; for the Olimex probe replace `openocd-jlink-rpi3.cfg` by `openocd-olimex.cfg`.

4.3 `aarch32-rpi-zero-2w` — hardware

```
D=$BUILD/aarch32-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test
```

`cmsis-os-validator` (allow about 300 s)

```
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-rpi3.cfg \  
-c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"  
sleep 12  
cat > /tmp/session.tcl <<'EOF'  
echo "--- stage: halt cores 0 1 2 3 ---"  
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }  
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"  
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }  
echo "--- stage: load_image __ELF__ ---"  
targets bcm2837.cpu0  
load_image __ELF__  
echo "--- stage: zero __smp_spin at 0x0003d400 (4 words) ---"  
mww 0x3d400 0  
mww 0x3d404 0  
mww 0x3d408 0  
mww 0x3d40c 0  
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"  
foreach core {0 1 2 3} {  
    targets [format {bcm2837.cpu%d} $core]  
    reg cpsr 0x600001da  
    resume 0x1003c  
}  
EOF  
sed -i "s|__ELF__|$D/cmsis-os-validator-hwd|g" /tmp/session.tcl  
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-rpi3.cfg -f /tmp/session.tcl
```

`mutex-stress` (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0002bfc0 (4 words) ---"
mww 0x2bfc0 0
mww 0x2bfc4 0
mww 0x2bfc8 0
mww 0x2bfcc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/mutex-stress-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

rtos-apis (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00051b68 (4 words) ---"
mww 0x51b68 0
mww 0x51b6c 0
mww 0x51b70 0
mww 0x51b74 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/rtos-apis-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

`sd_test` (allow about 450 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0004a070 (4 words) ---"
mww 0x4a070 0
mww 0x4a074 0
mww 0x4a078 0
mww 0x4a07c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/sd_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-mat-sdcard-test (allow about 900 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000e76c0 (4 words) ---"
mww 0xe76c0 0
mww 0xe76c4 0
mww 0xe76c8 0
mww 0xe76cc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-mat-sdcard-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-mat-test (allow about 900 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0006fca0 (4 words) ---"
mww 0x6fca0 0
mww 0x6fca4 0
mww 0x6fca8 0
mww 0x6fcac 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-mat-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-num-test (allow about 600 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00092630 (4 words) ---"
mww 0x92630 0
mww 0x92634 0
mww 0x92638 0
mww 0x9263c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-num-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-pipeline-test (allow about 600 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00083f30 (4 words) ---"
mww 0x83f30 0
mww 0x83f34 0
mww 0x83f38 0
mww 0x83f3c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-pipeline-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-pro-cons-test (allow about 600 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000412d0 (4 words) ---"
mww 0x412d0 0
mww 0x412d4 0
mww 0x412d8 0
mww 0x412dc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-pro-cons-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

`smp_test0` (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0002ce80 (4 words) ---"
mww 0x2ce80 0
mww 0x2ce84 0
mww 0x2ce88 0
mww 0x2ce8c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test0-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test1 (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0003c3f0 (4 words) ---"
mww 0x3c3f0 0
mww 0x3c3f4 0
mww 0x3c3f8 0
mww 0x3c3fc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test1-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test2 (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00040680 (4 words) ---"
mww 0x40680 0
mww 0x40684 0
mww 0x40688 0
mww 0x4068c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test2-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test3 (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0003dc30 (4 words) ---"
mww 0x3dc30 0
mww 0x3dc34 0
mww 0x3dc38 0
mww 0x3dc3c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test3-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test4 (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00054490 (4 words) ---"
mww 0x54490 0
mww 0x54494 0
mww 0x54498 0
mww 0x5449c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test4-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

usb_test (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00161848 (4 words) ---"
mww 0x161848 0
mww 0x16184c 0
mww 0x161850 0
mww 0x161854 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/usb_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

4.4 aarch32-rpi3b — hardware

```
D=$BUILD/aarch32-rpi3b-cmake-gcc-debug/platform-bin/port-tests/test
```

`ctest` registers no hardware case for `usb_test` on the Pi 3 B; its image is built but not listed here.

`cmsis-os-validator` (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0003d400 (4 words) ---"
mww 0x3d400 0
mww 0x3d404 0
mww 0x3d408 0
mww 0x3d40c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/cmsis-os-validator-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

mutex-stress (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0002bfc0 (4 words) ---"
mww 0x2bfc0 0
mww 0x2bfc4 0
mww 0x2bfc8 0
mww 0x2bfcc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/mutex-stress-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

rtos-apis (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00051b68 (4 words) ---"
mww 0x51b68 0
mww 0x51b6c 0
mww 0x51b70 0
mww 0x51b74 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/rtos-apis-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

`sd_test` (allow about 450 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0004a070 (4 words) ---"
mww 0x4a070 0
mww 0x4a074 0
mww 0x4a078 0
mww 0x4a07c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/sd_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-mat-sdcard-test (allow about 900 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000e7b10 (4 words) ---"
mww 0xe7b10 0
mww 0xe7b14 0
mww 0xe7b18 0
mww 0xe7b1c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-mat-sdcard-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-mat-test (allow about 900 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00070310 (4 words) ---"
mww 0x70310 0
mww 0x70314 0
mww 0x70318 0
mww 0x7031c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-mat-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-num-test (allow about 600 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000929b0 (4 words) ---"
mww 0x929b0 0
mww 0x929b4 0
mww 0x929b8 0
mww 0x929bc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-num-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-pipeline-test (allow about 600 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00084330 (4 words) ---"
mww 0x84330 0
mww 0x84334 0
mww 0x84338 0
mww 0x8433c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-pipeline-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp-pro-cons-test (allow about 600 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00041810 (4 words) ---"
mww 0x41810 0
mww 0x41814 0
mww 0x41818 0
mww 0x4181c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp-pro-cons-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

`smp_test0` (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0002ce80 (4 words) ---"
mww 0x2ce80 0
mww 0x2ce84 0
mww 0x2ce88 0
mww 0x2ce8c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test0-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

`smp_test1` (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0003c780 (4 words) ---"
mww 0x3c780 0
mww 0x3c784 0
mww 0x3c788 0
mww 0x3c78c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test1-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test2 (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00040680 (4 words) ---"
mww 0x40680 0
mww 0x40684 0
mww 0x40688 0
mww 0x4068c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test2-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test3 (allow about 120 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0003dfc0 (4 words) ---"
mww 0x3dfc0 0
mww 0x3dfc4 0
mww 0x3dfc8 0
mww 0x3dfcc 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test3-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

smp_test4 (allow about 300 s)

```

$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00054810 (4 words) ---"
mww 0x54810 0
mww 0x54814 0
mww 0x54818 0
mww 0x5481c 0
echo "--- stage: resume cores 0 1 2 3 at 0x1003c ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg cpsr 0x600001da
    resume 0x1003c
}
EOF
sed -i "s|__ELF__|$D/smp_test4-hwd|g" /tmp/session.tcl
$OPENOCD -s $A32/test/boards/rpi-zero-2w -s $SCRIPTS -f $A32/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -f /tmp/session.tcl

```

4.5 aarch64-rpi-zero-2w — hardware

```
D=$BUILD/aarch64-rpi-zero-2w-cmake-gcc-debug/platform-bin/port-tests/test
```

cmsis-os-validator (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000b9000 (8 words) ---"
mww 0xb9000 0
mww 0xb9004 0
mww 0xb9008 0
mww 0xb900c 0
mww 0xb9010 0
mww 0xb9014 0
mww 0xb9018 0
mww 0xb901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|${D}/cmsis-os-validator-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

mutex-stress (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000a7000 (8 words) ---"
mww 0xa7000 0
mww 0xa7004 0
mww 0xa7008 0
mww 0xa700c 0
mww 0xa7010 0
mww 0xa7014 0
mww 0xa7018 0
mww 0xa701c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/mutex-stress-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

rtos-apis (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000cd000 (8 words) ---"
mww 0xcd000 0
mww 0xcd004 0
mww 0xcd008 0
mww 0xcd00c 0
mww 0xcd010 0
mww 0xcd014 0
mww 0xcd018 0
mww 0xcd01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/rtos-apis-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

sd_test (allow about 450 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000e1000 (8 words) ---"
mww 0xe1000 0
mww 0xe1004 0
mww 0xe1008 0
mww 0xe100c 0
mww 0xe1010 0
mww 0xe1014 0
mww 0xe1018 0
mww 0xe101c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/sd_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-mat-sdcard-test (allow about 900 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0000000000172000 (8 words) ---"
mww 0x172000 0
mww 0x172004 0
mww 0x172008 0
mww 0x17200c 0
mww 0x172010 0
mww 0x172014 0
mww 0x172018 0
mww 0x17201c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|${D}/smp-mat-sdcard-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-mat-test (allow about 900 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000fa000 (8 words) ---"
mww 0xfa000 0
mww 0xfa004 0
mww 0xfa008 0
mww 0xfa00c 0
mww 0xfa010 0
mww 0xfa014 0
mww 0xfa018 0
mww 0xfa01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-mat-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-num-test (allow about 600 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0000000000012f000 (8 words) ---"
mww 0x12f000 0
mww 0x12f004 0
mww 0x12f008 0
mww 0x12f00c 0
mww 0x12f010 0
mww 0x12f014 0
mww 0x12f018 0
mww 0x12f01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|${D}/smp-num-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-pipeline-test (allow about 600 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0000000000139000 (8 words) ---"
mww 0x139000 0
mww 0x139004 0
mww 0x139008 0
mww 0x13900c 0
mww 0x139010 0
mww 0x139014 0
mww 0x139018 0
mww 0x13901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-pipeline-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-pro-cons-test (allow about 600 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000c3000 (8 words) ---"
mww 0xc3000 0
mww 0xc3004 0
mww 0xc3008 0
mww 0xc300c 0
mww 0xc3010 0
mww 0xc3014 0
mww 0xc3018 0
mww 0xc301c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-pro-cons-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

`smp_test0` (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000ac000 (8 words) ---"
mww 0xac000 0
mww 0xac004 0
mww 0xac008 0
mww 0xac00c 0
mww 0xac010 0
mww 0xac014 0
mww 0xac018 0
mww 0xac01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test0-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test1 (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000c9000 (8 words) ---"
mww 0xc9000 0
mww 0xc9004 0
mww 0xc9008 0
mww 0xc900c 0
mww 0xc9010 0
mww 0xc9014 0
mww 0xc9018 0
mww 0xc901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test1-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test2 (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000d1000 (8 words) ---"
mww 0xd1000 0
mww 0xd1004 0
mww 0xd1008 0
mww 0xd100c 0
mww 0xd1010 0
mww 0xd1014 0
mww 0xd1018 0
mww 0xd101c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "  start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test2-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test3 (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000ca000 (8 words) ---"
mww 0xca000 0
mww 0xca004 0
mww 0xca008 0
mww 0xca00c 0
mww 0xca010 0
mww 0xca014 0
mww 0xca018 0
mww 0xca01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "  start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test3-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test4 (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000f9000 (8 words) ---"
mww 0xf9000 0
mww 0xf9004 0
mww 0xf9008 0
mww 0xf900c 0
mww 0xf9010 0
mww 0xf9014 0
mww 0xf9018 0
mww 0xf901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test4-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

usb_test (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000001eb000 (8 words) ---"
mww 0x1eb000 0
mww 0x1eb004 0
mww 0x1eb008 0
mww 0x1eb00c 0
mww 0x1eb010 0
mww 0x1eb014 0
mww 0x1eb018 0
mww 0x1eb01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/usb_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

4.6 aarch64-rpi3b — hardware

```
D=$BUILD/aarch64-rpi3b-cmake-gcc-debug/platform-bin/port-tests/test
```

`ctest` registers no hardware case for `usb_test` on the Pi 3 B; its image is built but not listed here.

`cmsis-os-validator` (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000b9000 (8 words) ---"
mww 0xb9000 0
mww 0xb9004 0
mww 0xb9008 0
mww 0xb900c 0
mww 0xb9010 0
mww 0xb9014 0
mww 0xb9018 0
mww 0xb901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|${D}/cmsis-os-validator-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

mutex-stress (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000a7000 (8 words) ---"
mww 0xa7000 0
mww 0xa7004 0
mww 0xa7008 0
mww 0xa700c 0
mww 0xa7010 0
mww 0xa7014 0
mww 0xa7018 0
mww 0xa701c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/mutex-stress-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

rtos-apis (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000cd000 (8 words) ---"
mww 0xcd000 0
mww 0xcd004 0
mww 0xcd008 0
mww 0xcd00c 0
mww 0xcd010 0
mww 0xcd014 0
mww 0xcd018 0
mww 0xcd01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "  start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/rtos-apis-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

sd_test (allow about 450 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000e1000 (8 words) ---"
mww 0xe1000 0
mww 0xe1004 0
mww 0xe1008 0
mww 0xe100c 0
mww 0xe1010 0
mww 0xe1014 0
mww 0xe1018 0
mww 0xe101c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/sd_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-mat-sdcard-test (allow about 900 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0000000000172000 (8 words) ---"
mww 0x172000 0
mww 0x172004 0
mww 0x172008 0
mww 0x17200c 0
mww 0x172010 0
mww 0x172014 0
mww 0x172018 0
mww 0x17201c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-mat-sdcard-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-mat-test (allow about 900 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000fa000 (8 words) ---"
mww 0xfa000 0
mww 0xfa004 0
mww 0xfa008 0
mww 0xfa00c 0
mww 0xfa010 0
mww 0xfa014 0
mww 0xfa018 0
mww 0xfa01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-mat-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-num-test (allow about 600 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x0000000000012f000 (8 words) ---"
mww 0x12f000 0
mww 0x12f004 0
mww 0x12f008 0
mww 0x12f00c 0
mww 0x12f010 0
mww 0x12f014 0
mww 0x12f018 0
mww 0x12f01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|${D}/smp-num-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-pipeline-test (allow about 600 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000139000 (8 words) ---"
mww 0x139000 0
mww 0x139004 0
mww 0x139008 0
mww 0x13900c 0
mww 0x139010 0
mww 0x139014 0
mww 0x139018 0
mww 0x13901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-pipeline-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp-pro-cons-test (allow about 600 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000c3000 (8 words) ---"
mww 0xc3000 0
mww 0xc3004 0
mww 0xc3008 0
mww 0xc300c 0
mww 0xc3010 0
mww 0xc3014 0
mww 0xc3018 0
mww 0xc301c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp-pro-cons-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

`smp_test0` (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000ac000 (8 words) ---"
mww 0xac000 0
mww 0xac004 0
mww 0xac008 0
mww 0xac00c 0
mww 0xac010 0
mww 0xac014 0
mww 0xac018 0
mww 0xac01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test0-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

`smp_test1` (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000c9000 (8 words) ---"
mww 0xc9000 0
mww 0xc9004 0
mww 0xc9008 0
mww 0xc900c 0
mww 0xc9010 0
mww 0xc9014 0
mww 0xc9018 0
mww 0xc901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test1-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test2 (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000d1000 (8 words) ---"
mww 0xd1000 0
mww 0xd1004 0
mww 0xd1008 0
mww 0xd100c 0
mww 0xd1010 0
mww 0xd1014 0
mww 0xd1018 0
mww 0xd101c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "  start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test2-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test3 (allow about 120 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x000000000000cb000 (8 words) ---"
mww 0xcb000 0
mww 0xcb004 0
mww 0xcb008 0
mww 0xcb00c 0
mww 0xcb010 0
mww 0xcb014 0
mww 0xcb018 0
mww 0xcb01c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test3-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

smp_test4 (allow about 300 s)

```

$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init \
    -c "targets bcm2837.cpu0; halt; mww 0x3f100024 0x5a000001; mww 0x3f10001c 0x5a000020; shutdown"
sleep 12
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: halt cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 1 2 3 ---"
foreach core {0 1 2 3} { targets [format {bcm2837.cpu%d} $core]; arm semihosting enable }
echo "--- stage: load_image __ELF__ ---"
targets bcm2837.cpu0
load_image __ELF__
echo "--- stage: zero __smp_spin at 0x00000000000f9000 (8 words) ---"
mww 0xf9000 0
mww 0xf9004 0
mww 0xf9008 0
mww 0xf900c 0
mww 0xf9010 0
mww 0xf9014 0
mww 0xf9018 0
mww 0xf901c 0
echo "--- stage: resume cores 0 1 2 3 at 0x80000 ---"
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    reg pc 0x80000
    echo "    start PC core $core = [reg pc]"
}
foreach core {0 1 2 3} {
    targets [format {bcm2837.cpu%d} $core]
    resume
}
EOF
sed -i "s|__ELF__|$D/smp_test4-hwd|g" /tmp/session.tcl
$OPENOCD -s $A64/test/boards/rpi-zero-2w -s $SCRIPTS -f $A64/test/boards/rpi-zero-2w/openocd-jlink-
rpi3.cfg -c init -f /tmp/session.tcl

```

4.7 aarch32-luckfox-lyra — hardware

The Luckfox Lyra (RK3506, three Cortex-A7) is reached over SWD by a CMSIS-DAP probe (`$A32/test/boards/luckfox-lyra/openocd.cfg`), which runs `init` itself). Only core 0 is a debug target at load time — the image releases the other two — so only core 0 is halted, loaded and resumed, and there are no `__smp_spin` words to zero. Before the load, the MMU and caches the miniloader left on are switched off. There is no reset step. The entry read from every Lyra ELF is `0x20003c`.

```
D=$BUILD/aarch32-luckfox-lyra-cmake-gcc-debug/platform-bin/port-tests/test
CFG=$A32/test/boards/luckfox-lyra
```

mutex-stress-test (allow about 300 s)

```
cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [$_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]] { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/mutex-stress-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl
```

sd_test (allow about 450 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/sd_test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp-mat-sdcard-test (allow about 900 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp-mat-sdcard-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp-mat-test (allow about 900 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp-mat-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp-num-test (allow about 600 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp-num-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp-pipeline-test (allow about 600 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [$_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp-pipeline-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp-pro-cons-test (allow about 600 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp-pro-cons-test-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test0 (allow about 120 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test0-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test1 (allow about 120 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test1-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test2 (allow about 120 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]]
    } _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test2-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test3 (allow about 120 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test3-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test4 (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test4-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test5 (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test5-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test6 (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test6-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

smp_test7 (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test7-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

`smp_test_int` (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test_int-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

`smp_test_int2` (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test_int2-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

`smp_test_int3` (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test_int3-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

`smp_test_int4` (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {$_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test_int4-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

`smp_test_int5` (allow about 300 s)

```

cat > /tmp/session.tcl <<'EOF'
echo "--- stage: JTAG clock -> 4000 kHz ---"
adapter speed 4000
echo "--- stage: halt cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; halt }
echo "--- stage: enable semihosting on cores 0 ---"
foreach core {0} { targets [format {rk3506.a7.%d} $core]; arm semihosting enable }
echo "--- stage: pre-load ---"
targets rk3506.a7.0
if {[catch {
    set _t [target current]
    set _s [_t arm mrc 15 0 1 0 0]
    $_t arm mcr 15 0 1 0 0 [expr {_s & ~0x1005}]
    $_t arm mcr 15 0 7 5 0 0
    $_t arm mcr 15 0 7 5 6 0
    $_t arm mcr 15 0 8 7 0 0
} _e]} { echo "SANITIZE-ERR: $_e" } else { echo [format "SANITIZE-OK: SCTLR 0x%08x -> MMU/caches
off" $_s] }
echo "--- stage: load_image __ELF__ ---"
targets rk3506.a7.0
load_image __ELF__
echo "--- stage: resume cores 0 at 0x20003c ---"
foreach core {0} {
    targets [format {rk3506.a7.%d} $core]
    resume 0x20003c
}
EOF
sed -i "s|__ELF__|$D/smp_test_int5-hwd|g" /tmp/session.tcl
$OPENOCD -s $CFG -s $SCRIPTS -f $CFG/openocd.cfg -f /tmp/session.tcl

```

5. The same tests through `ctest`

The commands above are what these run:

```

cd $BUILD/<platform>-cmake-gcc-debug
ctest -N                                # the test names
ctest -R '^<platform>-<test>-qemu$' -V   # one QEMU test (run-qemu.sh)
ctest -L qemu -V                        # all the QEMU tests
ctest -R '^<platform>-<test>-hwd$' -V    # one hardware test (the board's hw.sh)

```

The board `hw.sh` looks for the shared runner in `micro-os-plus-iii-smp` (or `micro-os-plus-iii-smp.git`) next to the port. When the kernel folder is named `micro-os-plus-iii`, as here, it fails with

`.../micro-os-plus-iii-smp.git/test_smpl/run-hw.sh: No such file or directory` unless `UOS_SMP_DIR` is set:

```
UOS_SMP_DIR=$K ctest -R '^aarch32-rpi3b-smp_test0-hwd$' -V
```

`ctest` calls `hw.sh <test> 600`, so every hardware case gets 600 s; the per-test times in section 4 are the ones `run-hw.sh` uses when no time is given.